DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
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Project Synopsis

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Project Title: Safe-Stride Assist: An AI-Based Safe Walking Assistance System for People with Visual Impairments

Abstract:

Safe-Stride Assist is a research and teaching prototype that uses computer vision, monocular depth estimation, spatial reasoning, and speech feedback to support safer walking for people with visual impairment. A camera stream is processed locally using YOLO-based obstacle detection, Byte-Track tracking, and Depth Anything V2. The central contribution is a distance-aware walking-corridor algorithm that estimates the ground plane, converts relative depth into approximate meters, identifies the nearest obstacle or unknown blocker within a three-meter forward corridor, and recommends a clear side when possible. The system provides a browser-based live console, directional voice alerts, obstacle tracking, approach-risk estimation, and a visual path overlay. By combining named obstacle detection with depth-based blocker analysis, the prototype can provide actionable feedback instead of displaying detection boxes alone. Safe-Stride Assist is intended as an educational research prototype and not as a certified mobility, medical, or emergency device.

Keywords: Safe Walking Assistant, Computer Vision, YOLO, Monocular Depth Estimation, Distance Estimation, Obstacle Detection, Blocker Detection, Voice Alerts, Spatial Reasoning.

Introduction:

People with visual impairment may need to combine a white cane, guide dog, personal knowledge, and assistance from other people to interpret obstacles and move through unfamiliar environments. A camera-based assistant cannot replace these safety methods, but it can provide an additional perception channel by describing relevant blockers, estimating distance, and presenting timely audio guidance. Many computer vision demonstrations stop at detecting an object and drawing a bounding box. For safe walking assistance, the more important questions are whether a blocker lies in the user's forward path, how far away it is, whether it is approaching, and what action should be taken. Safe-Stride Assist addresses these questions using a modular local-processing pipeline that combines visual detection, depth estimation, geometric reasoning, risk scoring, and concise speech feedback.

Safe movement in an unfamiliar environment depends on understanding the relationship between the user, the ground, and nearby obstacles. A person, pole, door, vehicle, step, or other obstruction may be visible in the camera image but still be irrelevant if it is far to the side. Conversely, an unknown structure may be dangerous even when it is not represented by a detector class. This makes path-aware analysis important: the assistant must identify which visual information is close enough, high enough, and positioned within the user's forward walking corridor to deserve attention. Safe-Stride Assist is therefore designed as an additional perception and feedback layer rather than a replacement for a white cane, guide dog, or human support. The system uses a camera stream to estimate the scene, a depth model to approximate distance, and a spatial reasoning module to identify possible blockers in front of the user. Its output is deliberately short and action-oriented, for example reporting that the path is blocked, giving an approximate distance, and suggesting a clearer side. This approach connects computer vision results to a practical decision while keeping the user aware that the measurements are approximate.

Problem Statement:

Visually impaired individuals may encounter various obstacles while walking, including pedestrians, vehicles, poles, doorways, stairs, uneven surfaces, construction materials, and unfamiliar structures. However, conventional object-detection systems primarily identify predefined object classes without determining whether an obstacle poses an immediate risk, how far away it is, whether it is approaching, or what action should be taken. While obstacles located to the side may not present an immediate threat, unrecognized objects directly ahead can create significant safety hazards. To address these limitations, the proposed system aims to develop an assistive safe-walking prototype that integrates known-object detection, monocular depth estimation, ground-plane distance correction, three-meter walking-corridor analysis, unknown-obstacle detection, approach-risk assessment, and prioritized voice guidance. The system will determine whether a detected object or an unfamiliar obstruction lies within the user's forward walking path, estimate its approximate distance, identify its relative direction, assess its potential risk, and provide concise audio instructions, such as stopping or moving towards a clearer side when appropriate. Additionally, prioritized voice alerts will help minimize unnecessary announcements and ensure that critical warnings receive greater attention. Since factors such as camera orientation, lighting conditions, motion blur, ground visibility, and scene complexity can affect detection and distance estimation accuracy, the system will provide approximate guidance rather than guaranteed measurements. It is intended to serve as a supplementary mobility aid.

Literature Survey:

Redmon et al. [1] introduced the YOLO family of one-stage object detectors, demonstrating that visual detection can be performed at high speed by processing the complete image in a single network. This supports the real-time detection component of Safe-Stride Assist, although the project uses a current Ultralytics YOLO checkpoint and filters the output to walking-relevant classes. **Zhang et al. [2]** proposed Byte-Track, a multi-object tracking method that associates high- and low-confidence detections to preserve object identities across frames. Temporal tracking is important for Safe-Stride Assist because persistence and approach speed help reduce one-frame false alerts and support time-to-collision reasoning. **Yang et al. [3]** introduced Depth Anything V2 for monocular depth estimation. The model supplies relative scene depth from a single camera frame. Safe-Stride Assist extends this capability with a ground-plane estimator and camera-height assumption to obtain a stable approximate metric scale for nearby walking hazards. **Ranftl et al. [4]** demonstrated the usefulness of monocular depth models for dense scene understanding. Their work highlights that relative depth captures scene structure but does not automatically provide a guaranteed real-world distance. This motivates the ground-plane scale correction and explicit safety limitations used in the proposed prototype. **World Health Organization [5]** reports the importance of accessible support for people with vision impairment. Assistive systems should therefore communicate only relevant information, use understandable feedback, and avoid presenting experimental computer vision as a replacement for established mobility aids. These principles guide Safe-Stride Assist's prioritized alerting and prototype safety boundary. Research on assistive computer vision also shows that perception quality and interaction design must be evaluated together. A detector may achieve good image-level accuracy but still be unsuitable if it speaks too frequently, gives incorrect side instruction, or reports a distant object that does not affect the walking path. Safe-Stride Assist therefore treats distance, path relevance, and alert timing as first-class parts of the system rather than as optional display features. **Khan et al. [7]** systematically reviewed navigation assistants for blind and visually impaired people and identified continuing gaps in independent, reliable, and context-aware navigation. This supports Safe-Stride Assist's combination of perception, route guidance, and safety feedback rather than treating obstacle detection as an isolated task. **Xu et al. [8]** reviewed wearable electronic travel aids for obstacle avoidance. Their findings support combining camera-based perception with distance estimation and multimodal feedback, while also reinforcing that such systems should complement conventional mobility aids. **Katzschmann et al. [9]** demonstrated local navigation using time-of-flight sensing and haptic feedback. This provides direct support for distance-aware local hazard warnings and for offering feedback channels that do not depend only on speech.

Eltaher et al. [10] reviewed urban navigation systems for visually impaired people, including route guidance and location-aware assistance. Their work supports Safe-Stride Assist's destination search, route instructions, heading alignment, off-route detection, and arrival state. **Johannesson et al. [11]** presented the Sound of Vision project, which explored converting visual scene information into auditory representations. This supports the use of concise directional voice feedback and spatial audio as a non-visual interface, while Safe-Stride Assist keeps its output intentionally shorter and action-oriented. **Shull and Damian [12]** reviewed haptic wearables as sensory replacement and sensory augmentation. Their analysis supports the future addition of haptic cues and the use of multimodal feedback when speech alone may be insufficient or distracting. **Ranganeni et al. [13]** studied different levels of control in a navigation assistant for blind travelers. Their results indicate that user preferences vary by situation, supporting Safe-Stride Assist's role as an assistive layer that provides guidance while preserving user agency rather than claiming full autonomous control. **Zhang et al. [14]** proposed an ARCore-based user-centric assistive navigation system for visually impaired people. This supports the feasibility of combining smartphone cameras, computer vision, positioning, and user-directed guidance in a portable application context. The literature further indicates that monocular depth should be treated as an estimate with scene-dependent uncertainty. Camera height, pitch, ground visibility, lighting, motion blur, and reflective surfaces affect the result. The proposed system addresses this by fitting a ground plane from lower-image points, smoothing scale estimates over recent frames, using a bounded forward corridor, and communicating the prototype's limitations instead of presenting distance as a guaranteed measurement.

Existing System & its Drawback:

Existing camera-based assistance demonstrations commonly use object detection to draw labels and bounding boxes over a live image. Some systems add monocular depth or tracking, but these components are often presented independently rather than combined into a single walking-path decision. Generic detectors also cannot identify every real-world hazard, and a detector label alone does not indicate whether an obstacle blocks the user's path or how urgently it should be announced.

Drawbacks of Existing Systems:

- **Limited Reach of Manual Aids:** Khan et al. [7] and the reviewed navigation literature explain that electronic travel aids are intended to complement, rather than replace, traditional mobility aids. A white cane or guide dog remains essential, but manual aids may provide limited information about distant, overhead, or unfamiliar obstacles.
- **Limited Distance Awareness:** Xu et al. [8] show that obstacle-avoidance aids must consider sensing range and feedback, because a visual label or bounding box alone does not indicate how urgently the user must respond.
- **No Walking-Corridor Reasoning:** Existing detection systems commonly identify objects in the camera view without deciding whether they lie within the user's forward walking corridor. This can produce warnings for objects that are visible but not directly relevant to the user's path.
- **Unknown Blockers May Be Missed:** Assistive vision systems depend strongly on their sensing and recognition models. Temporary structures, low-contrast hazards, and objects without trained classes may therefore remain undetected.
- **Limited Ground-Change Awareness:** General obstacle detectors may not reliably distinguish every curb, step, or change in ground level from ordinary scene structure. This remains a secondary extension area for Safe-Stride Assist rather than the main focus of the current prototype.
- **Repeated and Low-Value Alerts:** Frequent announcements can increase cognitive load and make urgent information harder to notice. A system must therefore consider persistence, distance, path relevance, and alert priority rather than speaking for every detection.
- **Weak Temporal Reasoning:** Frame-by-frame analysis does not establish whether an obstacle is approaching, stationary, or moving across the walking path. Tracking methods such as Byte-Track [2] are needed to provide more stable temporal evidence.

- **Additional Hardware and Connectivity Requirements:** Some assistive solutions require dedicated sensors, wearable components, or network services, which can increase cost, maintenance, and dependence on operating conditions.
- **Unclear Safety Boundaries:** Ranftl et al. [4] show that monocular depth is useful for scene structure but does not automatically provide guaranteed real-world distance. Experimental systems must communicate uncertainty and must not be treated as replacements for certified mobility support.

Proposed Methodology:

The proposed Safe-Stride Assist system is a local, AI-based safe-walking assistant that processes camera frames and generates concise voice guidance. It combines **YOLO obstacle detection, Byte-Track temporal tracking, Depth Anything V2 monocular depth estimation, RANSAC ground-plane fitting**, a three-dimensional walking-corridor model, approach-risk estimation, and prioritized speech feedback. The system can detect known obstacles and can also identify unknown blockers from depth points that rise above the estimated ground plane within the forward corridor. The distance algorithm begins by converting the depth map into three-dimensional points using a pinhole-camera model and an estimated focal length. A RANSAC procedure searches the lower image for a stable horizontal ground plane while rejecting points belonging to people, walls, vehicles, and other outliers. The known approximate camera height is then used to estimate a scale factor. Recent valid scale estimates are retained in a short history and combined using a median, reducing sudden distance changes caused by individual noisy frames.

After scale correction, the spatial analyzer tests each point against a one-meter-wide corridor extending from approximately 0.8 meters to three meters in front of the camera. Points that rise above the ground plane within the configured obstacle-height range are treated as possible blockers. If enough valid points are present, the nearest robust depth percentile is selected as the obstacle distance. The distribution of front-facing points is then used to estimate clock direction and whether the left or right side appears clearer. The alert decision layer combines corridor status with named obstacle detections, tracking persistence, estimated approach speed, and time-to-collision. A nearby path blocker receives higher priority than a distant obstacle outside the corridor. The announcer waits briefly for persistence to remove one-frame false positives, prevents messages from interrupting each other unnecessarily, and repeats an alert only after a cooldown or meaningful reduction in distance. This produces short messages such as 'Stop. Path blocked, 2 meters. Keep left.'

The key features of the proposed system include:

1. **Real-Time Obstacle Detection:** Uses YOLO to identify walking-relevant obstacles from webcam or video frames.
2. **Temporal Obstacle Tracking:** Uses Byte-Track to maintain obstacle identity and support persistence and approach analysis.
3. **Monocular Distance Estimation:** Uses Depth Anything V2 and ground-plane scale correction to estimate approximate distance in meters.
4. **Walking-Corridor Analysis:** Evaluates a one-meter-wide corridor up to three meters ahead of the camera.
5. **Unknown Blocker Detection:** Detects depth points that rise from the ground even when no named obstacle class is available.
6. **Actionable Voice Alerts:** Produces messages containing obstacle type, distance, direction, urgency, and recommended side.
7. **Local Browser Console:** Displays the annotated feed, path state, depth status, alert priority, captions, and diagnostics.
8. **Safety-Oriented Alerting:** Uses persistence, cooldowns, urgency, and path priority so that the most relevant warning is spoken first.

Objectives:

The primary objectives of the proposed system are:

- To develop a real-time computer vision pipeline for safe-walking assistance using a camera stream.
- To detect walking-relevant obstacles using YOLO and track them over time with Byte-Track.
- To estimate approximate obstacle distance using monocular depth and ground-plane calibration.
- To analyze a three-meter forward corridor and determine whether it is blocked.
- To detect unknown depth-based blockers that do not have a recognized obstacle label.
- To generate concise voice alerts containing obstacle distance, direction, urgency, and guidance.
- To reduce false and repeated alerts using persistence checks, cooldowns, and priority ranking.
- To provide a browser console for monitoring the live feed, path status, and alert state.
- To establish a foundation for future custom hazard and environmental-state models.

Proposed System Advantages:

- **Distance-Aware Assistance:** Converts relative monocular depth into an approximate distance that can be included in speech feedback, supporting more informative warnings than object labels alone, as demonstrated by Yang et al. [3] and Ranftl et al. [4].
- **Known and Unknown Blocker Reasoning:** Combines named YOLO obstacles with depth-based corridor analysis to identify possible hazards even when they do not belong to a trained object class, building on the object-detection and depth-estimation approaches of Redmon et al. [1], Yang et al. [3].
- **Walking-Path Relevance:** Estimates whether an obstacle lies inside the user's forward walking corridor instead of treating every detected region equally. This supports the local-navigation objectives discussed by Katzschmann et al. [9] and the navigation systems reviewed by Eltaher et al. [10].
- **Actionable Feedback:** Provides a recommended clear side when the estimated scene geometry supports a left or right correction. Directional audio and haptic feedback have also been explored by Katzschmann et al. [9], Johannesson et al. [11], and Shull and Damian [12].
- **Reduced Alert Overload:** Uses detection persistence, urgency, cooldowns, approach risk, and priority ranking to keep speech concise and meaningful. This follows the user-centred navigation and control considerations discussed by Khan et al. [7] and Ranganeni et al. [13].
- **Real-Time Demonstration:** Supports webcam and recorded-video processing with an annotated browser console. The real-time detection and temporal tracking components are supported by Redmon et al. [1] and Zhang et al. [2].
- **Modular and Extensible Design:** Separates detection, depth estimation, spatial reasoning, risk analysis, and speech feedback so that future hazard models and interaction methods can be evaluated independently, consistent with the modular directions identified in the reviews by Khan et al. [7], Xu et al. [8], and Eltaher et al. [10].
- **Clear Safety Boundary:** Reports the system as an educational research prototype with approximate distance rather than a certified mobility device. This is consistent with the assistive-technology limitations and complementary-use principles discussed by Khan et al. [7] and Xu et al. [8].
- **Local and Privacy-Oriented Processing:** Processes the camera stream locally by default and provides browser-based feedback without requiring continuous remote human assistance. This creates a practical alternative to systems that depend heavily on network connectivity or external operators [7], [8].

Software Requirements:

SOFTWARE	SPECIFICATION
OPERATING SYSTEM	Windows 11 / Ubuntu 22.04 LTS
PROGRAMMING LANGUAGE	Python 3.10 or higher
COMPUTER VISION LIBRARY	OpenCV
OBJECT DETECTION MODEL	YOLO11s using Ultralytics
OBJECT TRACKING	Byte-Track
DEEP LEARNING FRAMEWORK	PyTorch
DEPTH ESTIMATION MODEL	Depth Anything V2
WEB INTERFACE	HTML, CSS, JavaScript, and browser speech synthesis

Hardware Requirements:

HARDWARE	SPECIFICATION
PROCESSOR	Intel Core i7
GRAPHICS PROCESSING UNIT (GPU)	NVIDIA GPU with CUDA Support
RAM	8 GB Minimum
STORAGE	256 GB SSD or Higher
CAMERA	IP webcam, Digital Camera
AUDIO OUTPUT	Speaker or headphones for voice alerts

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